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Understanding impacts of aggressive driving on freeway safety and mobility: A multi-agent driving simulation approach

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ABSTRACT

Considering that aggressive driving behavior significantly contributes to crash occurrences, it is necessary to establish appropriate safety strategies for managing and suppressing the risk factor of aggressive driving behavior. Understanding the characteristics of aggressive driving and its impact on traffic stream is fundamental to developing safety countermeasures. The main purpose of this study is to evaluate the crash potential under the various aggressive driving events on freeways based on both driving simulators and microscopic traffic simulation model, VISSIM. A novel feature of this study is the application of a multi-agent driving simulation facility, where two driving simulators are connected to establish a network that can render synchronized driving in the same space. The results of the driving simulation were used for modifying the driving behavior parameters of VISSIM. This approach will enable to effectively represent the driving behavior of aggressive drivers and normal drivers under various aggressive driving cases. Study results indicated that aggressive driving deteriorated not only the network safety performance represented by the crash potential index (CPI) but also the mobility represented by travel speeds. The outcomes of this study would be useful in developing well-informed countermeasures and supporting policy making-activities to prevent aggressive driving events.

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1. Introduction

Driving behavior is a crucial factor affecting road traffic safety. In particular, it is well known that aggressive driving behavior is closely related to the occurrence and severity of crashes (Clapp et al., 2011; Stephens & Sullman, 2014; Sullman, 2015; Benavente et al., 2007). The NCHRP Report 500 (Neuman et al., 2003) defined aggressive driving as operating a motor vehicle in a selfish, pushy, or impatient manner, often unsafely, directly affecting other drivers. In addition, Dula and Geller (2003) reported that aggressive driving consists of any type of behavior directed hurting another person without taking into consideration those person's rights or safety. Following closely, performing abrupt and risky lane changes, car ramming, and speeding are in the category of aggressive driving events that deteriorate stable interactions with nearby vehicles.

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Therefore, understanding the characteristics of aggressive driving and its impact on traffic stream in terms of safety performance is fundamental to developing countermeasures to suppress aggressive driving and eventually achieve the goals of crash prevention.

Attempts to understand the characteristics of aggressive driving behavior have been widely made from a perspective of the psychological point of view. Bogdan, Măirean, and Havârneanu (2016) conducted a useful meta-analysis to identify the relationship based on a thorough review of 51 eligible studies. From a different point of view, analyzing driving behavior as a hazardous event leading to the crash occurrence has been widely addressed by researchers in the field of traffic engineering. Moghaddam, Tabibi, Sadeghi, Ayati, and Ghotbi Ravandi (2017) investigated the relationship between the driver characteristics and the abnormal driving behavior leading to crashes. The proposed risk index can be used to assess crash risks based on driving behavior. Eboli, Mazzulla, and Pungillo (2016) developed a methodology to determine whether a driver is driving safely based on the analysis of the road segment length, speed, and lateral acceleration data obtained from smart phones in the field. Moreover, driving simulation experiments were conducted to evaluate the driver's perception of risks (Takahashi et al., 2017). The driver's palm sweat response, skin reaction, steering, accelerator and brake operating ability were used to identify the driver's perceived risk. Another branch of earlier studies addressed how to detect aggressive driving events (Carboni & Bogorny, 2015; Li, Miyajima, Kitaoka, & Takeda, 2014; Murphey, Milton, & Kiliaris, 2009). Those studies attempted to develop certain criteria based on deriving critical values to identify and characterize aggressiveness while driving. However, fewer efforts have been made in quantifying the crash potential of aggressive driving that would be necessary to derive effective countermeasures. Moreover, identifying whether aggressive driving would give an impact on the mobility performance is of interest. These research opportunities motivate our study.

The purpose of this study is to evaluate the crash potential of aggressive driving events on freeways. To date, we are aware of several studies that conducted the crash risk evaluation of risky and aggressive driving (Habtemichael & de Picado Santos, 2013, 2014; Jeong, Oh, Lee, & Cho, 2014) based on VISSIM microscopic traffic simulations. Compared to these existing works, a novel feature of our study is to use both driving and traffic simulations. In particular, this study uses a multi-agent driving simulation environment for better understanding of vehicle interactions between aggressive driving vehicles and their surrounding vehicles. Two driving simulators are connected to establish a network environment capable of simulating driving in the same space synchronized through a multi-agent software. The primary purpose of using a multi-agent driving simulator is to obtain more reliable behavior data representing aggressive driving by injecting a normal vehicle being interacting with the aggressive vehicle. It is more realistic for a subject to move in a situation where there exists an adjacent partner vehicle, rather than simply moving along. The results of driving simulations are then used for modifying the driving behavior parameters of VISSIM to effectively represent the driving behavior of aggressive drivers and normal drivers. To quantify crash risks, the crash potential index (CPI) using deceleration to avoid crash (DRAC) and maximum available deceleration rate (MADR), by Cunto and Saccomanno (2008), is applied in our analysis. In addition, the impact of different levels of the composition rate (CR) of aggressive driving vehicles on the crash potential is further analyzed. The freeway mobility evaluation using the average travel speed is conducted as well as safety assessment using the CPI.

The remainder of this paper is organized as follows. Section 2 presents the proposed methodology, including the proposed comprehensive simulation framework consisting of driving and traffic simulations. Section 3 presents the analysis results along with discussions. Finally, Section 4 presents the conclusion with a summary of this study and future research directions based on the identified limitations of this study.

2. Methodology

2.1. Overall framework

The proposed analysis methodology consists of three stages: (1) multi-agent driving simulations, (2) microscopic traffic simulations, and (3) CRI evaluations. A multi-agent driving simulation is conducted to identify the characteristics of vehicle interactions among an aggressive driving vehicle and surrounding normal vehicles affected by the aggressive vehicle. The main purpose of the first stage is to compare the driving behavior of normal driving and aggressive driving. The multi-agent driving simulation environment is this study will allow capturing various driving behavior patterns in a more scientific and systematic manner. Therefore, results of this stage would help identify the differences in driving behavior between normal and aggressive drivers as well as assess the extent of aggressive driving behavior impact on the normal driving pattern. The results of this stage were further processed to derive parameters for microscopic traffic simulations. In the second stage, a microscopic traffic simulator, VISSIM, was used to investigate the impact of aggressive driving events on the network performance. The driving behavior parameters of VISSIM were modified based on the results of multi-agent driving simulations in Stage 1. Different CR levels of aggressive driving vehicles were injected into the traffic simulation, and the resulting changes in vehicle trajectories were obtained. The change in mobility performance due to aggressive driving events was evaluated by comparing average travel speeds. In the last stage, CPI was used to quantify crash risks caused by the aggressive driving and ultimately to evaluate the safety performance of traffic stream. Fig. 1 illustrate overall framework proposed in this study. More technical details for each stage are presented in the subsequent sections.

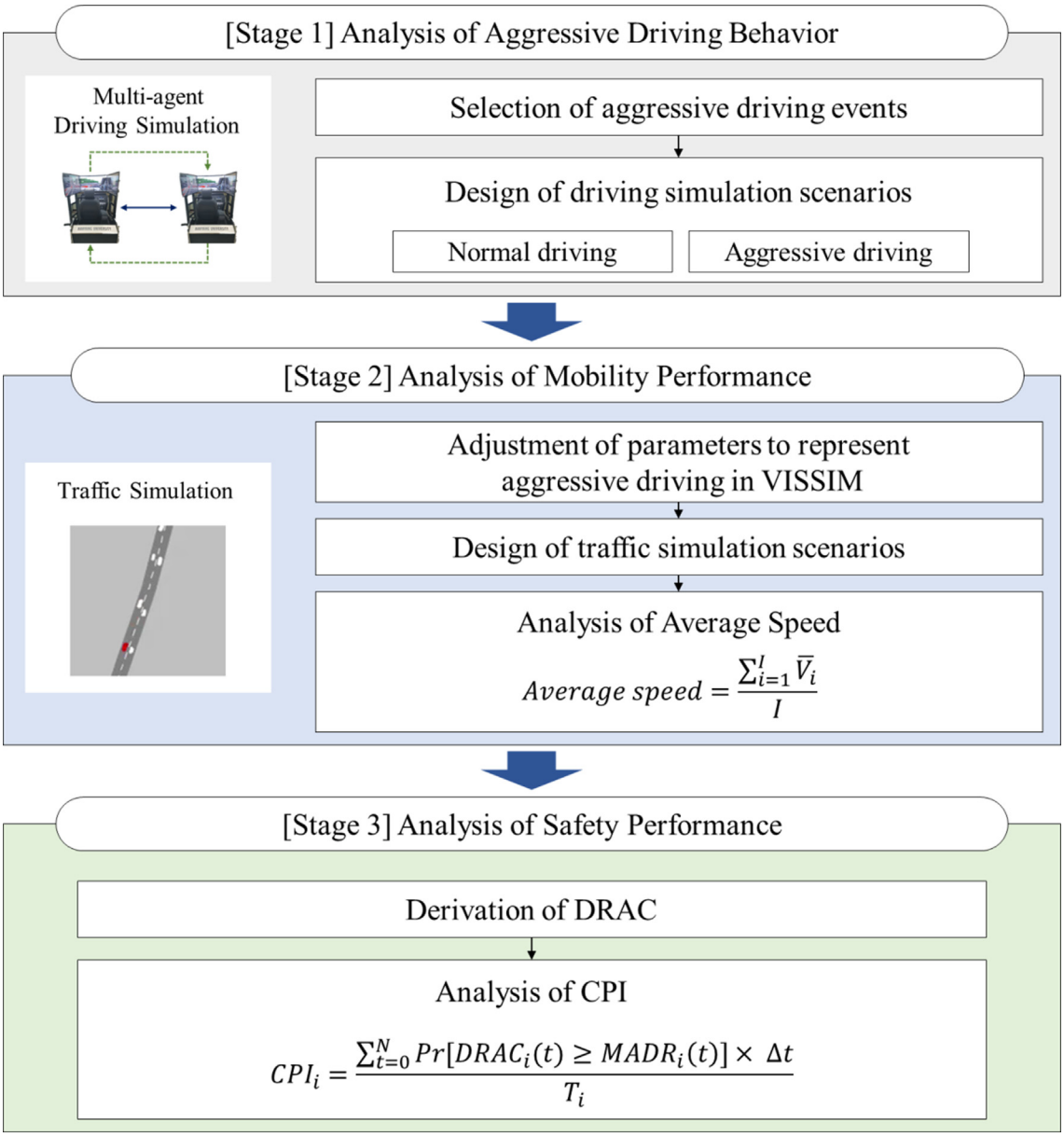


Fig. 1. Overall framework.

2.2. Multi-agent driving simulation

In this study, a multi-driver simulation environment is constructed that can simulate driving in the same road space synchronized by connecting two driving simulators. Two simulator computers are connected through TCP/IP (Transmission Control Protocol/Internet Protocol) to exchange driving data, such as position, speed, acceleration, and angular velocity of the aggressive driving vehicle and a normal vehicle affected by aggressive driving. In the simulated environment, two participants were traveling at the same time, with one participant playing the role of the aggressive driver and the other person playing the role of the general driver affected. More realistic aggressive driving behavior is obtainable in the multi-agent driving simulation environment due to the existence of an adjacent vehicle being interacting with the aggressive vehicle. The concept of multi-agent driving simulation is shown in Fig. 2.

This study classified the simulated road into three sections such as before passing, passing, and after passing. Participants were instructed to conduct three aggressive driving events including following closely, abrupt lane change, and sudden deceleration on before passing, passing, and after passing sections, respectively. Fig. 3 illustrates the characteristics of aggressive driving events in this study.

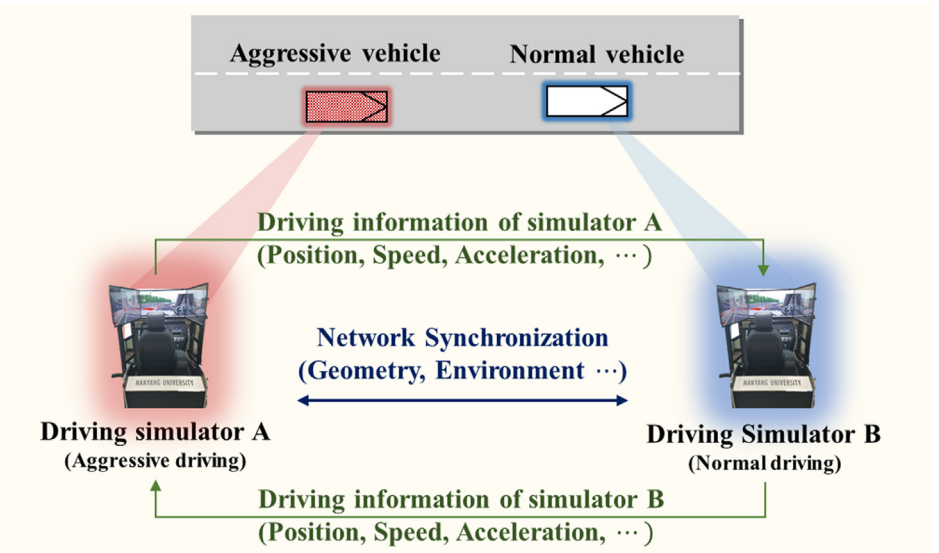


Fig. 2. Concept of multi-agent driving simulation.

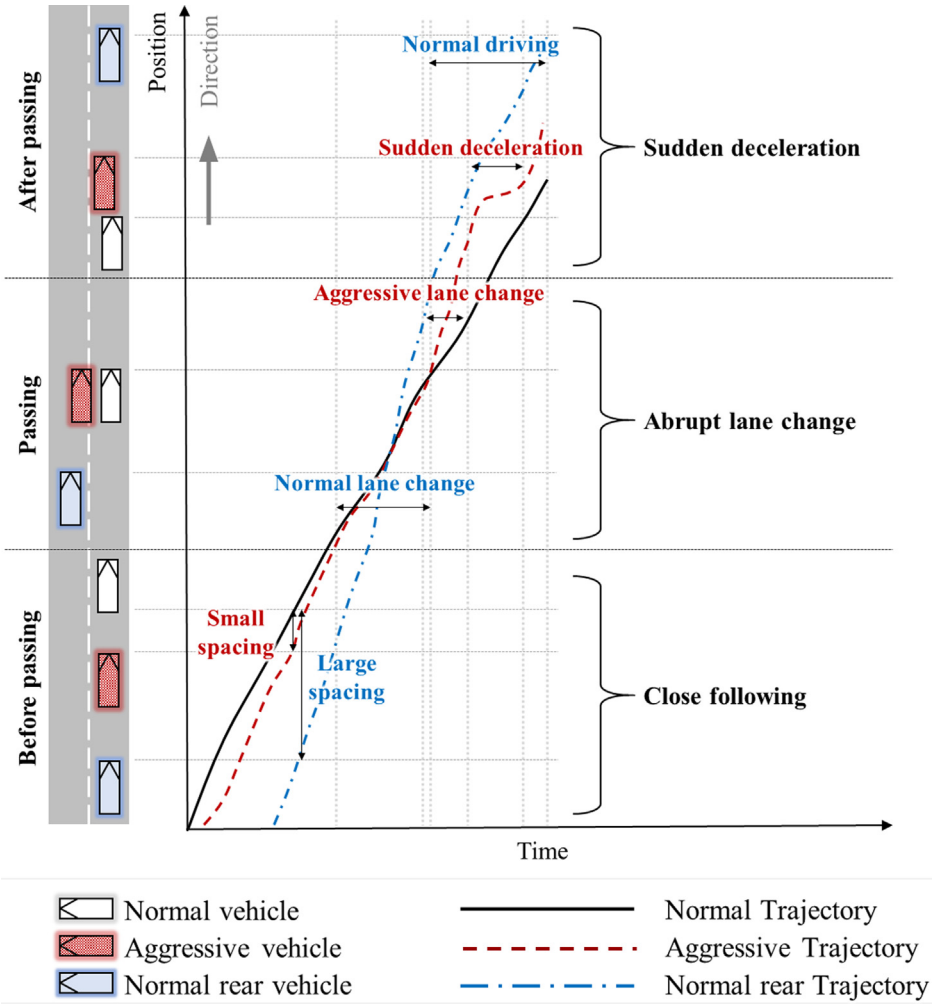


Fig. 3. Characteristics of aggressive driving events.

2.3. Determination of behavior parameters for VISSIM

This study focused on three aggressive driving events: (1) following closely, (2) abrupt lane change, and (3) sudden deceleration, each of which can be implemented by the driving simulation experiment. The results obtained from the multi-agent driving simulation experiments were further processed to be coded for modifying behavior parameters for VISSIM (PTV Group, 2015). The following closely behavior involves ignoring the safety distance in a car-following situation and threatening the front vehicle. Taking into consideration short spacing between the aggressive vehicle and the front vehicle, parameters associated with spacing and headway were selected to simulate aggressive driving vehicles performing the following closely. To simulate vehicles performing the abrupt lane change with a shorter gap and a higher speed, parameters that affect the lane change behavior were selected. In addition, acceleration related parameters were modified to represent the sudden deceleration event. As a result, a total of 11 parameters were selected to represent the behavior of aggressive driving vehicles in view of the aforementioned situations. The selected parameters, as shown in Table 1, can be classified into three groups that represent vehicle maneuvering: as longitudinal, lateral, and acceleration.

Because the driving simulation environment has a limitation in reproducing the reality completely, it is undesirable to give significant meaning to the absolute value of the data obtained from the experimental results. Therefore, it can be said that results should be used to determine the priority or importance of scenarios by giving more significant meaning to the relative evaluation. Under this circumstance, a viable approach is to use the relative ratios of the experimental data obtained from both the normal driving and the aggressive driving cases, rather than directly applying the resultant absolute values representing driving behavior. In this study, the parameter value to model aggressive behavior in VISSIM was adjusted based on the corresponding ratio value for each parameter. In addition, it should be noted that ten simulation runs for each scenario were conducted with different seed numbers to consider the stochastic nature of varying results, which potentially represent the different behavioral consequences due to the various drivers' responses to some extent.

2.4. Evaluation of crash risk

The CPI (Cunto & Saccomanno, 2008) was used in this study to evaluate the impact of aggressive driving events on traffic stream in terms of traffic safety. Two measures representing a deceleration condition that would lead to the crash occurrence are required to estimate the CPI. Those measures include the deceleration to avoid crash (DRAC) presented in Eq. (1) and the maximum available deceleration rate (MADR) representing the braking capability of the subject vehicle. Regarding applying the MADR, Cunto (2008) proposed different MADRs for vehicle types, such as cars and trucks, based on the assumption of following truncated normal distributions. Because our driving simulation experiments are limited to cars, we used an average of 8.45 m/s^2 with a standard deviation of 1.40 m/s^2 for MADR. The CPI is defined as the probability that the DRAC of the subject vehicle exceeds the MADR when the subject vehicle follows a leading vehicle. To evaluate the impact of aggressive driving behavior on the safety performance of traffic stream by the CPI, vehicle trajectory data were extracted from the VISSIM simulation model by taking the different levels of MPR of aggressive vehicles into consideration. The CPI for vehicle i can be given by Eq. (2):

$$DRAC = \frac{(V_L(t) - V_F(t))^2}{2(X_L(t) - X_F(t) - L_L)}, \quad V_F(t) > V_L(t) \tag{1}$$

$V_L(t)$: speed of leading vehicle at time t
$V_F(t)$: speed of following vehicle at time t

Table 1
Selected behavior parameters for simulating aggressive driving events.

VISSIM parameter		Parameter definition
Longitudinal	CC1 [s]	It is the distance in seconds which a driver wants to maintain at a certain speed.
	CC3	It controls the start of the deceleration process.
	CC7 [m/s^2]	Oscillation during acceleration.
	Desired speed distribution [km/h]	If not hindered by other vehicles, a driver will travel at his desired speed (with a small stochastic variation called oscillation).
Lateral	Look ahead distance [m]	The distance that a vehicle can see forward in order to react to other vehicles either in front or to side of it.
	Min. headway [m]	Minimum distance to the vehicle in front that must be available for a lane change in standstill condition.
	Safety distance reduction factor	During lane changes the reduction factor is regarded which takes effect for the safety distance of the trailing vehicle in the new lane for the decision whether to change lanes or not.
	Minimum time gap [s]	Between the conflict marker and the next approaching vehicle (merging).
	Minimum headway [m]	Between conflict marker and next vehicle upstream (merging).
Acceleration	Desired acceleration [m/s^2]	Used for any other situation.
	Desired deceleration [m/s^2]	If it is lower than the max. deceleration, then the desired deceleration is used as the maximum for deceleration.

$X_L(t)$: position of leading vehicle at time t
$X_F(t)$: position of following vehicle at time t
L_L: length of leading vehicle

$$\mathrm{CPI_i} = \frac{\sum_{t=0}^{N} Pr[DRAC_i(t) > MADR_i(t)] \times \Delta t}{T} \tag{2}$$

$MADR_i(t)$: MADR of the vehicle i at time step t
N: Total number of time intervals
Δt: Observation time interval
T: Total observation time period ($\mathrm{T} = \mathrm{N} \times \Delta \mathrm{t}$)

3. Analysis and results

3.1. Driving behavior analysis: characteristics of aggressive driving

A total of 32 subjects consisting of 28 males and 4 females in their 20 s with driving experience over 1 year on freeways participated in driving simulation experiments from August 14 to August 18 in 2017 at the driving behavior analysis laboratory of Hanayang University. A 2-kilometer (km) freeway segment with two-lane per direction was modeled using a 3D virtual reality design software, UC-win/Road. Each participant spent approximately 30 min for completing driving simulations.

The subject was paid a certain amount of hourly incentive. Subjects for normal driving were instructed to drive at a constant speed without making any lane changes. We did not provide subjects with any information that the following vehicle would perform aggressive driving, which leads to the natural reaction of the subject. With regard to the instruction for aggressive driving, subjects were asked to participate in the experiment while recalling the experience of feeling anger on a real road in the past. In addition, we explained to the subjects how to perform specific aggressive driving events defined in this study. It is expected that the existence and reaction of an adjacent vehicle of the aggressive vehicle are useful in obtaining more naturalistic and realistic aggressive driving events.

The driving simulation scenario consisted of general driving scenarios and aggressive driving scenarios. The participants for the normal driving scenario were asked to travel at their comfortable speeds in a range not exceeding the speed limit. More specifically, in the normal driving scenario, a driver (subject 1, preceding vehicle) started driving on the right lane and was instructed to drive without any lane change. The subject 2 (follower) started driving 200 m behind the subject 1. It was then instructed to follow the preceding vehicle while maintaining a safe distance up to the first 1 km in the entire 2 km-long simulated road section. When the experimental assistant instructed the participant to change lane, the participant conducted a lane change at a safe and comfort mode. In contrast, the participants for the aggressive driving scenario were asked to travel as aggressive as possible. Under the aggressive driving scenario, the participant was required to follow vehicles closely, abruptly perform a lane change, and perform sudden deceleration after passing. Participants were asked to conduct such aggressive driving events in order. As expected, small spacing and abrupt change in speeds were observed. To evaluate whether simulated aggressive driving events are sufficiently different from normal driving, four measurements including acceleration, angular velocity, steering, and spacing were analyzed using the *t*-test. Results show that there exist statistically significant differences between aggressive and normal driving behavior at 95% confidence level as presented in Table 2.

3.2. VISSIM simulation: network calibration

A 16.7-km freeway stretch between the Nakdong and Seonsan interchanges on Korean freeway 45 was selected as the study site (Fig. 4). This four-lane freeway section has a rest area in one direction and has several tunnels with climbing lanes. According to a recent crash statistics report (Statistical Analysis on Traffic Accidents, 2014–2016), this study section has been identified as one of the most crash-prone locations in the Korean freeway system.

This freeway section used in this study was coded in a VISSIM microscopic traffic simulator and calibrated in a previous study conducted by a research team from the Center for Traffic Information and Safety at Hanyang University, Korea (Song & Oh, 2011; Jeong and Oh, 2017). Parameters for car-following and lane-changing models were adjusted to minimize the difference between actual data and simulated data. 'u-statistic' presented in Eq. (3) was used to determine the acceptability of simulation results. In general, u-statistic close to 0 represents that actual and simulated data are similar. In case u-statistic is less than 0.1, calibration results can be accepted (Brockfeld et al., 2004). The 15-min interval loop detector data collected at 12 points in the real field were used for the simulation calibration. The simulation calibration was focused on representative traffic conditions. We analyzed the effectiveness of systems under various traffic environments by changing the traffic flow characteristics for each scenario. Calibration results are summarized in Table 3.

Table 2
Comparison of vehicle maneuvering by driving behavior.

Section	Driving event	Measurement	Behavior		t-statistic (p-value)
			Aggressive (Avg/Std)	Normal (Avg/Std)	
Before passing	Rear-close following	Acceleration [m/s^2]	0.7190 (0.6690)	0.4090 (0.3720)	−2.294 (0.025)
		Angular velocity [rad/s]	0.0006 (0.0009)	0.0003 (0.0003)	−2.030 (0.047)
		Steering* [n/a]	0.0008 (0.0008)	0.0005 (0.0005)	−1.885 (0.064)
		Spacing [m]	34.2053 (27.3807)	64.0738 (32.8883)	3.948 (0.000)
Passing	Sudden path change	Acceleration [m/s^2]	0.6480 (0.3200)	0.4050 (0.1650)	−3.829 (0.000)
		Angular velocity [rad/s]	0.0058 (0.0030)	0.0033 (0.0016)	−4.123 (0.000)
		Steering* [n/a]	0.0071 (0.0035)	0.0042 (0.0021)	−4.069 (0.000)
After passing	Sudden deceleration	Acceleration [m/s^2]	1.7330 (0.7390)	0.2770 (0.2110)	−10.711 (0.000)
		Angular velocity [rad/s]	0.0049 (0.0052)	0.0024 (0.0011)	−2.644 (0.010)
		Steering* [n/a]	0.0057 (0.0064)	0.0034 (0.0014)	−2.001 (0.050)
		Spacing [m]	29.7051 (17.3965)	78.9943 (33.0964)	7.457 (0.000)

* Steering: A numerical expression of the relative steering angle, which ranged from −1 to + 1, with a negative value representing turning the steering wheel to the left and a positive value representing turning the steering wheel to the right.

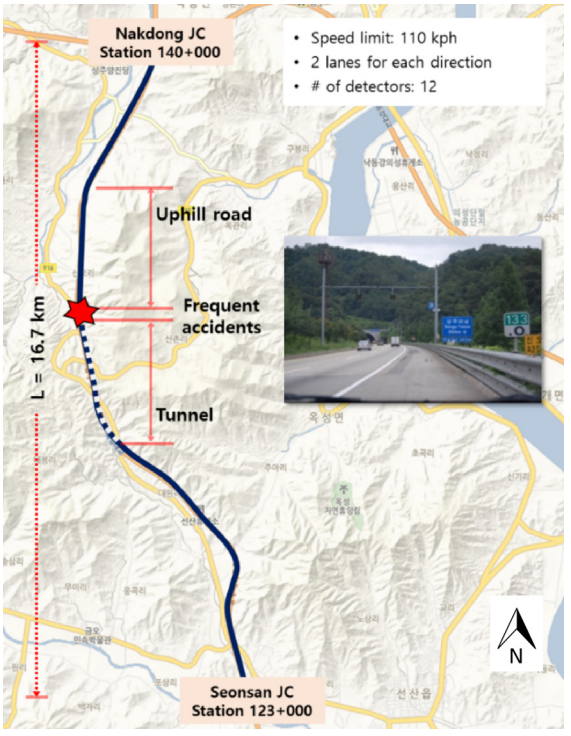


Fig. 4. Study area.

Table 3
Comparison of u-values between field and simulation loop detector data.

Detector ID	1	2	3	4	5	6	7	8	9	10	11	12	Average
u-value	0.016	0.021	0.038	0.028	0.021	0.030	0.030	0.025	0.028	0.021	0.032	0.013	0.025

$$U = \frac{\sqrt{\sum_{i=1}^{N}\left(V_i - \widehat{V_i}\right)^2}}{\sqrt{\sum_{i=1}^{N} V_i^2} - \sqrt{\sum_{i=1}^{N} \widehat{V_i}^2}} \quad (3)$$

were V_i = actual speed (kph); $\widehat{V_i}$ = simulation speed (kph)

3.3. VISSIM simulation: analysis and parameter determination

Driving behavior data obtained from three aggressive events including following closely on the before passing section, abrupt lane change on the passing section, and sudden deceleration on the after passing section were analyzed to be used for adjusting behavior parameters of VISSIM. The simulation results showed that the average vehicle spacing is 62.2 m when the following vehicle is normal, whereas the aggressive vehicle has a much shorter spacing of 32.4 m, which is approximately half the spacing. This finding was used to determine a parameter value of 'Minimum headway (m)'. The minimum headway for aggressive driving was adjusted to be two times smaller than that of the normal driving; the normal minimum headway is a default value in VISSIM. The obtained headway value was further processed to derive 'minimum time gap (sec)' based on the relationship between spacing and speed. It was identified that the minimum time gaps for the aggressive and normal cases were 0.4 sec and 0.7 sec, respectively. Therefore, the parameter values for both CC1 and minimum time gap for the aggressive driving were set 0.68 times lower than the default value. Regarding the acceleration, the average of the acceleration values of aggressive driving is approximately 4.5 times larger than that of the normal driving. For simulating aggressive driving, this result was applied to adjust the CC7 value, which is 4.5 times higher than the default value to represent the normal driving behavior. Meanwhile, speeds and accelerations were input to the traffic simulation in the form of distribution to consider participants' various responses. More specifically, 15th percentile, 50th percentile, 85th percentile, and the maximum value were used for the parameter values of 'Desired speed distribution'. In addition, the distributions of acceleration and deceleration by speeds obtained from the simulation were used for determining 'desired acceleration and desired deceleration. Because the parameter values for CC3, 'Look ahead distance', and 'Safety distance reduction factor' cannot be obtained from the driving simulation experiments, we adopted the parameter values recommended by an existing study (Habtemichael & de Picado Santos, 2014). Table 4 presents parameter values to be used for evaluating the impact of aggressive driving on traffic stream using VISSIM.

3.4. Crash risk analysis

Various MPR levels of aggressive driving vehicles contributing to traffic safety were systematically investigated as presented in Fig. 5. The CR was increased by 10% increment starting from 10% till it reaches the maximum value of 50%. Regarding traffic conditions, this study simulated the situation in which the traffic volume corresponding to the level of service (LOS) D is generated in VISSIM as a representative scenario. LOS D conditions represent a moderate level of traffic situation in which there is no traffic congestion and the vehicle's freedom of movement is not very high. It was observed that the average CPI increases as the CR of aggressive vehicles increases. When the CR was 10%, the CPI increased approximately 1.6 times that of the case of 0% CR, which represented normal traffic conditions without aggressive driving events. Although the CR of 50% is somewhat exaggerated, the average CPI was 3.6 times higher than that of 0% CR.

The analysis of variance (ANOVA) test was further performed to assess whether the CPI values for each MPR level had statistically significant differences. The null hypothesis noting that CPIs of each CR scenario were not different was rejected at the significance level of 0.05. Additionally, a post hoc test was performed to test the feasibility of grouping different CR scenarios when the CPI difference among reviewed scenarios was weak. Based on the post hoc results, five CR scenarios

Table 4
Comparison of normal and aggressive driving parameters.

VISSIM parameter		Normal driving	Aggressive driving
Longitudinal	CC1 [s]	0.9	0.52
	CC3	−8.00	−4.00*
	CC7 [m/s^2]	0.25	1.12
	Desired speed distribution [km/h]	74–105	75–140
Lateral	Look ahead distance [m]	250	150*
	Min. headway [m]	0.5	0.25
	Safety distance reduction factor	0.6	0.8*
	Minimum time gap [s]	3.0	1.7
	Minimum headway [m]	5.0	2.5
Acceleration	Desired acceleration [m/s^2]	0–3.5	0.2–5
	Desired deceleration [m/s^2]	−3.0 to −2.5	−10 to 0

* Adopt parameters presented by Habtemichael et al. (2014).

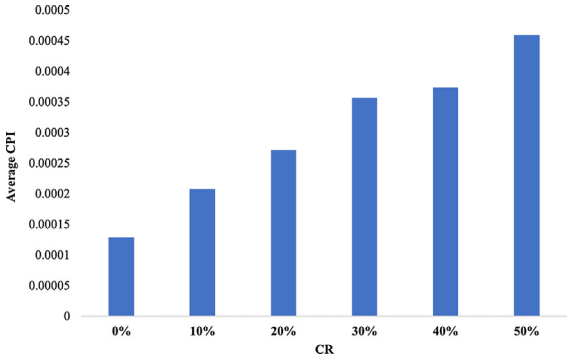


Fig. 5. Relationship between CPI and CR.

Table 5
Results of statistical tests.

Levene test for equality of variances			
Levene's test	Degree of freedom 1	Degree of freedom 2	p-value
0.611	5	54	0.692

One-Way ANOVA

	Sum of squares	**df**	**Mean square**	**F-statistics**	**Sig.**
Between Groups	0.000	5.000	0.000	52.252	0.000
Within Groups	0.000	54.000	0.000		
Total	0.000	59.000			

Post analysis

CR (%)		**N**	**Group(subset for alph = 0.05)**			
			1	**2**	**3**	**4**
Turkey HSDa	MPR 0	10	0.00013			
	MPR 10	10		0.00021		
	MPR 20	10		0.00027		
	MPR 30	10			0.00036	
	MPR 40	10			0.00037	
	MPR 50	10				0.00046
	p-value		1.0000	0.09093	0.97629	1.0000

in this study were clustered into three groups, (1) CR levels of 10% and 20% in one group, (2) CR levels of 30% and 40% as another one, and (3) CR 50% scenario as a standalone group. Table 5 describes the results of ANOVA and post hoc analyses.

3.5. Mobility analysis

In addition to the analysis of crash risks, the impact of aggressive driving on the freeway mobility was evaluated by comparing the average speeds of each CR scenario. As illustrated in Fig. 6, the average speed decreased with the increase of the CR

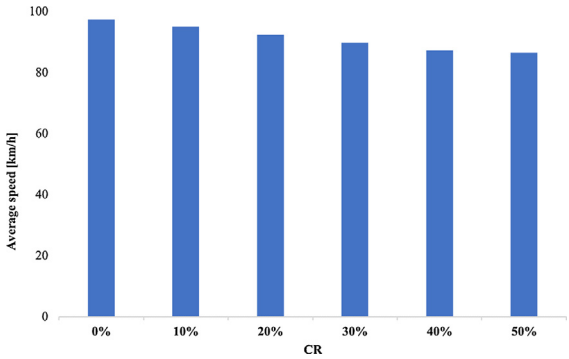
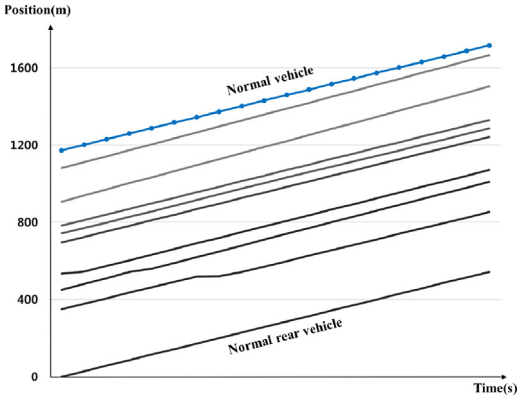
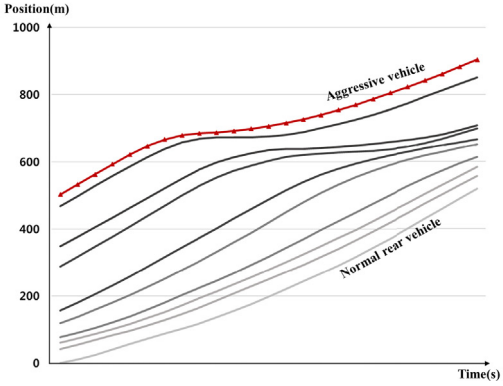


Fig. 6. Reduction of average speed by CR.



(a) Normal vehicle trajectories



(b) Aggressive vehicle trajectories

Fig. 7. Comparison of vehicle trajectories.

level of aggressive driving vehicles. This finding implies that aggressive driving behavior (e.g. following closely, abrupt lane change, and sudden deceleration) would negatively affect the mobility as well as the traffic safety. For example, in case of CR of 30% scenario, approximately 10% of speed reduction was observed when compared to the baseline condition (i.e. CR of 0% case). An in-depth analysis showed that frequent deceleration was observed among the vehicles following aggressive vehicles, thus resulting in developing shock waves. Fig. 7 illustrates the difference in the driving pattern of sample vehicle trajectories extracted from the VISSIM simulation. Unlike the aggressive driving, the variation of the vehicle trajectory for normal driving was not observed to a great extent. It is because we simulated continuous traffic flow on a freeway.

4. Conclusion

Aggressive driving behavior including following closely, abrupt lane change, and sudden deceleration would deteriorate stable interactions with nearby vehicles. Because unstable vehicle interactions are highly associated with the potential of crash occurrence, countermeasures to suppress the aggressive driving are expected to prevent crashes effectively. Understanding the impact of aggressive driving on traffic stream would be the first step toward the development of such countermeasures. Achieving this understanding motivated our study to quantify the crash potential caused by aggressive driving. It is expected that the proposed analysis framework would be effectively used for scientific and systematic understanding of the impact of driving behavior on the performance of traffic stream.

The proposed methodology to achieve the purpose of this study consisted of two simulation experiments. Both driving simulations to identify the characteristics of behavioral data and microscopic traffic simulations to evaluate the impact of aggressive driving on traffic stream were conducted. A multi-agent driving simulation facility in which driving simulators are connected to establish a network environment capable of simulating synchronized driving in the same space was used to obtain aggressive and normal driving behavior data, which is the outcome of vehicle interactions.

The driving characteristics of aggressive drivers were identified using a multi-agent driving simulation technique, and the aggressive behavior was implemented by adjusting the parameters of VISSIM for conducting microscopic traffic simulations. The CPI was adopted to quantify the crash potential and to evaluate the safety performance of each analysis scenario by

different levels of MPR of aggressive vehicles. The mobility evaluation using the average speed was conducted as well as safety assessment using the CPI. The result revealed that the crash potential increased with the increase of the MPR. It is meaningful that we quantified the magnitude of increase in the crash potential by analysis scenarios. Moreover, an interesting finding was that the increase in MPR results in degrading the mobility performance by the reduction of average travel speed. The outcomes of this study would be useful in developing promising countermeasures and supporting policy-making activities to prevent aggressive driving events. For example, these results can be used to develop education and campaign materials for aggressive driving. In addition, the effectiveness of advanced driving assistance systems (ADAS) and traffic management strategies can be assessed in terms of reducing aggressive driving events.

To further advance findings, the following future research should be conducted. First, in this study, we did not recruit sufficient participants to systematically generalize the characteristics of the driving behavior according to drivers' gender and age. As a further study, more experiments with a larger group of participants to meet the proper composition ratios including sufficient female and elderly drivers need to be conducted. Second, vehicle maneuvering parameters need to be further derived for identifying vehicle interactions between aggressive and normal driving vehicles. Those parameters can be taken into consideration in simulating aggressive vehicles in the microscopic traffic simulations in support of the VISSIM COM interface, rather than simply changing behavior parameter values. Finally, systematic and scientific analyses of data obtained from a naturalistic driving study should be conducted for deriving more reliable results.

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